Prompt-Only Frontier LLMs Beat Lightweight Music Recommendation Baselines on Public Ranking Tasks

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Abstract

We study a narrow but practical question: if a user exports music listening history and asks a frontier language model for recommendations, can prompting alone match or beat lightweight Spotify-style recommenders? We evaluate this question as closed-set ranking on two public tasks: LAST.FM next-track prediction and SPOTIFY MPD playlist continuation. Each test example contains one held-out relevant track and 19 negatives, so classical baselines and language models solve the same 20-way ranking problem. We compare two prompt-only LLMs, GPT-4.1 and CLAUDE SONNET 4.5, against two classical baselines: a popularity ranker and a personalized cooccurrence heuristic.

The strongest results come from the simplest prompt. On LAST.FM, GPT-4.1 with the MINIMAL prompt reaches 0.589 MRR and 0.621 NDCG@10, compared with 0.282 MRR and 0.246 NDCG@10 for the strongest classical baseline. On SPOTIFY MPD, the same setup reaches 0.329 MRR and 0.379 NDCG@10, compared with 0.094 MRR and 0.061 NDCG@10 for cooccurrence. These gains are statistically significant under paired Wilcoxon signed-rank tests with Benjamini-Hochberg correction. Adding compact profile hints does not help and is reliably worse on SPOTIFY MPD.

Our main conclusion is narrow but clear: prompt-only frontier LLMs can outperform lightweight history-based music recommendation baselines on public offline ranking tasks. At the same time, this result does not establish that prompting beats Spotify's production recommender, because we evaluate public datasets, fixed candidate pools, and simpler baselines than a production system is likely to use.

1 Introduction

Users can now export years of listening history from streaming platforms and hand that history directly to a general-purpose language model. That creates a simple question with real practical stakes: if we ask a frontier LLM for music recommendations from exported data, do we get recommendations that are as good as platform-style recommenders?

This question matters because prompt-only recommendation changes who controls the interface. Instead of relying entirely on a platform recommender, users could carry their history across systems and request recommendations in natural language. If prompting alone works well, recommendation becomes more portable, inspectable, and user-directed.

Existing work suggests that LLMs can help recommendation, but most strong results come from hybrid systems rather than pure prompting. Survey papers map a broad design space for LLM-enhanced recommendation, including prompting, instruction tuning, and hybrid pipelines Wu et al. [2023], Zhao et al. [2023]. Foundational work such as P5 and instruction-following recommendation reformulates recommendation as language processing Geng et al. [2022], Zhang et al. [2023]. More recent systems such as LLARA, A-LLMREC, and TALKPLAY combine LLMs with sequential recommenders, collaborative filtering, or multimodal music representations Liao et al. [2024], Kim et al. [2024], Doh et al. [2025]. What remains less clear is the narrower setting that many users actually care about: no fine-tuning, no learned retriever, and no private platform signals, only prompting over listening history.

We examine exactly that setting. We cast recommendation as closed-set ranking and compare prompt-only frontier LLMs against lightweight classical baselines on two public music tasks: LAST.FM next-track prediction and SPOTIFY MPD playlist continuation. Every method sees the same 20-item candidate set, with one held-out positive and 19 negatives. This setup avoids an unfair comparison between unconstrained generation and numerical ranking while preserving the central question of whether the model can identify the right continuation from context alone. Figure 1 summarizes the evaluation protocol.

The main quantitative result is simple. On LAST.FM, GPT-4.1 with the MINIMAL prompt reaches 0.589 MRR and 0.621 NDCG@10, compared with 0.282 and 0.246 for the strongest classical baseline. On SPOTIFY MPD, the same setup reaches 0.329 MRR and 0.379 NDCG@10, compared with 0.094 and 0.061. CLAUDE SONNET 4.5 is statistically tied with GPT-4.1 in the same prompt condition, and richer profile prompts do not improve results.

Our contributions are:

- We provide a controlled head-to-head evaluation of prompt-only frontier LLMs against lightweight Spotify-style baselines on two public music recommendation tasks.
- We show that the strongest prompt-only condition significantly outperforms both a popularity baseline and a personalized cooccurrence baseline on LAST.FM and SPOTIFY MPD.
- We conduct prompt ablations and statistical tests that show a simple recent-context prompt is as good as or better than a richer profile prompt.
- We clarify the boundary of the practical claim: these results support prompt-only recommendation against lightweight public baselines, not against Spotify's proprietary production recommender.

The rest of the paper is organized as follows. Section 2 reviews prior work on LLM-based and music-specific recommendation. Section 3 describes the evaluation setup, baselines, and statistical plan. Section 4 reports main results and ablations. Section 5 discusses interpretation, limitations, and implications.

2 Related Work

LLMs as recommendation models. Early work showed that recommendation tasks can be cast as language generation. P5 reframed multiple recommendation problems as text-to-text prediction Geng et al. [2022], while instruction-following recommendation used templated instructions and task-specific tuning to improve recommendation quality Zhang et al. [2023]. Survey papers argue that LLMs add semantic reasoning, flexible interfaces, and stronger zero-shot behavior, but they also emphasize unresolved issues around grounding, evaluation, and personalization Wu et al. [2023], Zhao et al. [2023]. Our work is narrower: we do not tune a model or train an adapter, and we focus on whether prompting alone is competitive on public music ranking tasks.

Prompting versus hybrid recommendation. Several recent systems suggest that LLMs are often strongest when paired with classical recommender components rather than used alone. LLM-Rec uses prompting to enrich item text for downstream recommenders Lyu et al. [2023]. LLARA aligns sequential recommender embeddings with LLM inputs Liao et al. [2024], and A-LLMREC injects collaborative knowledge from a frozen collaborative filter into a frozen LLM Kim et al. [2024]. These systems report strong gains, but they answer a different question from ours because they rely on learned recommenders, alignment modules, or training data beyond the prompt itself. We instead isolate the prompt-only regime.

Music recommendation and playlist continuation. Public music recommendation benchmarks remain centered on playlist continuation and next-item prediction. The SPOTIFY MPD challenge literature provides a strong non-LLM reference point for Spotify-style recommendation, including collaborative filtering, neighbor methods, and hybrid reranking systems Teinemaa et al. [2018], Zamani et al. [2018]. More recently, TALKPLAY shows that LLM-based music recommendation can work well when the model is trained end to end over multimodal music representations Doh et al. [2025]. Our work sits between these lines of research: unlike the playlist continuation literature, we evaluate frontier LLMs directly; unlike TALKPLAY, we do not train a music-specific model.

History or playlist prefix → 20-item candidate pool → method-specific ranking → offline metrics

One ground-truth continuation and 19 negatives are shared across all methods. LLMs receive the same candidate pool as the classical baselines and return an ordered ranking.

Figure 1: Evaluation pipeline. We test prompt-only recommendation as candidate ranking, not open-ended generation, so every method is compared on the same decision problem.

Popularity bias and beyond-accuracy evaluation. Recommendation quality is not only about top-1 or top-10 accuracy. Recent work studies how LLM recommenders interact with popularity bias and catalog concentration Lichtenberg et al. [2024]. That perspective is especially relevant for music, where users often value novelty and stylistic fit. Following this line of work, we report both ranking metrics and simple beyond-accuracy statistics, including mean top-10 popularity percentile and coverage.

Taken together, prior work shows that LLMs can contribute meaningfully to recommendation, especially in hybrid systems. The missing piece is a clean measurement of prompt-only recommendation quality in a music setting using the same candidate sets as classical baselines. That is the gap we target.

3 Methodology

Task formulation. Each evaluation example consists of context h, a candidate set $C = \{c_1, \dots, c_{20}\}$, and one relevant held-out target $y \in C$. The context is either a recent listening history (Last.fm) or a playlist prefix (Spotify MPD). Methods must rank the 20 candidates, and we score the induced ranking by MRR, NDCG@10, and Hit@10. This closed-set protocol ensures that LLMs and classical baselines solve the same problem.

3.1 Datasets and Example Construction

Last.fm next-track ranking. The local pre-gathered subset contained only seven users, so we loaded a larger Hugging Face slice, 'train[:1000000]', which yielded 56 users and 55 eligible users. We sampled 50 held-out next-track examples. Each example uses recent chronological listening history as context, one true next track as the positive item, and 19 negatives.

Spotify MPD playlist continuation. We used a downloaded slice of 1000 playlists from the Spotify Million Playlist Dataset and constructed 50 held-out continuation examples. Each example uses a playlist prefix as context and the same 1-positive, 19-negative candidate structure as Last.fm.

Why closed-set ranking? The user-facing question is often framed as open-ended recommendation, but direct comparison is otherwise ill-posed. A free-form LLM could generate items outside the baseline candidate pool, while a classical heuristic typically scores a fixed list. We therefore use shared candidate sets for all methods.

3.2 Methods

Classical baselines. We implement two lightweight baselines. Popularity ranks candidates by training frequency. Cooccurrence scores each candidate with a combination of track cooccurrence, artist cooccurrence, and a small popularity term. This is intended as a lightweight Spotify-style heuristic, not a reproduction of Spotify's production stack.

Prompt-only LLM conditions. We evaluate GPT-4.1 through the OpenAI Responses API and Claude Sonnet 4.5 through OpenRouter, both at temperature 0.0. We test two prompts. The Minimal prompt includes only recent listening or playlist context plus the candidate list. The Profile prompt adds compact profile hints such as top artists and country when available.

3.3 Metrics and Statistical Analysis

Ranking metrics. Our primary metrics are MRR and NDCG@10, which are well suited to single-target candidate ranking. We also report Hit@10. For beyond-accuracy analysis, we report Mean-

Method	MRR	NDCG@10	Hit@10	Mean Top-10 Popularity Pct
Popularity	0.051	0.000	0.000	0.0001
Cooccurrence	0.282	0.246	0.260	0.0100
GPT-4.1 (MINIMAL)	**0.589**	**0.621**	**0.780**	0.0388
GPT-4.1 (PROFILE)	0.539	0.581	**0.780**	**0.0398**
CLAUDE SONNET 4.5 (MINIMAL)	0.572	0.597	0.740	0.0361
CLAUDE SONNET 4.5 (PROFILE)	0.544	0.586	**0.780**	0.0397

Table 1: Results on LAST.FM next-track ranking. Higher is better for all metrics shown. Best results are in **bold**.

Method	MRR	NDCG@10	Hit@10	Mean Top-10 Popularity Pct
Popularity	0.067	0.036	0.100	0.0012
Cooccurrence	0.094	0.061	0.120	0.0494
GPT-4.1 (MINIMAL)	**0.329**	**0.379**	0.640	**0.0832**
GPT-4.1 (PROFILE)	0.203	0.206	0.380	0.0710
CLAUDE SONNET 4.5 (MINIMAL)	0.312	0.373	**0.660**	0.0805
CLAUDE SONNET 4.5 (PROFILE)	0.244	0.285	0.540	0.0747

Table 2: Results on SPOTIFY MPD playlist continuation. Higher is better for all metrics shown. Best results are in **bold**.

TOP10-POPULARITY-PCT, where larger values indicate that recommendations are less concentrated in the global popularity head, and coverage, defined as the fraction of unique catalog items appearing in top-10 recommendation lists across the test set.

Statistical tests. We run paired Wilcoxon signed-rank tests on per-example MRR and NDCG@10 differences and apply Benjamini-Hochberg correction across primary comparisons within each dataset. We report paired Cohen's d on per-example differences and bootstrap 95% confidence intervals for aggregate metrics.

3.4 Environment and Cost

Execution environment. Experiments were run on May 5, 2026 with Python 3.12.8 and standard scientific Python libraries, including NumPy 2.4.4, pandas 3.0.2, SciPy 1.17.1, matplotlib 3.10.9, seaborn 0.13.2, datasets 4.8.5, and requests 2.33.1. The machine exposed four NVIDIA RTX A6000 GPUs, but the evaluation was API-bound and did not use local GPU inference.

API cost. The final run used 84,195 OpenAI input tokens and 23,014 output tokens, for an estimated OpenAI cost of about $0.353 using the official GPT-4.1 API pricing page. Claude usage totaled 123,526 tokens with reported API cost of $0.606. Total estimated cost was about $0.958.

4 Results

4.1 Main Comparison Against Classical Baselines

LAST.FM. Table 1 shows that every LLM condition outperforms both classical baselines. The best configuration is GPT-4.1 with the MINIMAL prompt, which reaches 0.589 MRR and 0.621 NDCG@10. This is a large absolute gain over COOCCURRENCE at 0.282 MRR and 0.246 NDCG@10. The bootstrap 95% confidence interval for GPT-4.1 (MINIMAL) is [0.464, 0.711] for MRR and [0.508, 0.732] for NDCG@10, while COOCCURRENCE reaches [0.171, 0.396] and [0.140, 0.366].

SPOTIFY MPD. Table 2 shows the same pattern on playlist continuation. GPT-4.1 (MINIMAL) reaches 0.329 MRR and 0.379 NDCG@10, compared with 0.094 and 0.061 for COOCCURRENCE. The corresponding bootstrap 95% confidence intervals are [0.235, 0.434] and [0.277, 0.485] for GPT-4.1 (MINIMAL), versus [0.061, 0.142] and [0.016, 0.119] for COOCCURRENCE.

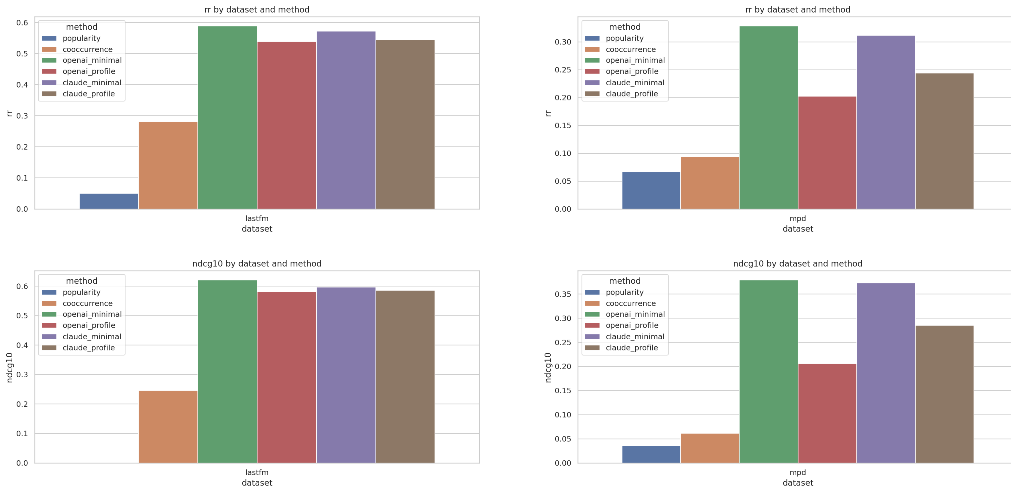

Figure 2: Ranking performance on both datasets. Left column: Last.fm reciprocal rank and NDCG@10 comparisons. Right column: Spotify MPD reciprocal rank and NDCG@10 comparisons. In both datasets, prompt-only LLM conditions separate clearly from the classical baselines, with the simplest prompt producing the strongest results.

4.2 Significance Tests

Primary comparisons. The gains over the strongest classical baseline remain significant after Benjamini-Hochberg correction on both datasets. On Last.fm, GPT-4.1 (Minimal) beats Cooccurrence with corrected $p = 4.69 \times 10^{-6}$ on MRR and $p = 1.44 \times 10^{-5}$ on NDCG@10, with paired effect sizes $d = 0.70$ and $d = 0.85$. Claude Sonnet 4.5 (Minimal) also beats Cooccurrence with corrected $p = 1.10 \times 10^{-5}$ on MRR and $p = 3.82 \times 10^{-5}$ on NDCG@10, with $d = 0.62$ and $d = 0.75$.

On Spotify MPD, GPT-4.1 (Minimal) beats Cooccurrence with corrected $p = 3.57 \times 10^{-7}$ on MRR and $p = 7.88 \times 10^{-6}$ on NDCG@10, with $d = 0.69$ and $d = 0.85$. Claude Sonnet 4.5 (Minimal) shows similarly strong gains, with corrected $p = 3.57 \times 10^{-7}$ on MRR and $p = 5.93 \times 10^{-6}$ on NDCG@10, and effect sizes $d = 0.72$ and $d = 0.93$.

4.3 Prompt Ablation and Model Comparison

Minimal prompts are enough. The richer Profile prompt does not improve performance. On Last.fm, the difference between Minimal and Profile is not significant for either model. On Spotify MPD, the profile prompt is reliably worse. For Claude Sonnet 4.5, Minimal beats Profile with $p = 0.00138$ on MRR and $p = 0.00990$ on NDCG@10. For GPT-4.1, Minimal beats Profile with $p = 0.00034$ on MRR, $p = 0.00114$ on NDCG@10, and $p = 0.00079$ on Hit@10.

Model comparison. The best GPT-4.1 and Claude Sonnet 4.5 prompt-only systems are statistically tied on both datasets. On Spotify MPD, for example, GPT-4.1 (Minimal) and Claude Sonnet 4.5 (Minimal) differ only by $p = 0.963$ on MRR and $p = 0.891$ on NDCG@10.

4.4 Beyond-Accuracy Behavior

Popularity concentration. The LLM conditions consistently achieve higher mean top-10 popularity percentiles than the classical baselines. On Last.fm, GPT-4.1 (Minimal) reaches 0.0388 versus 0.0100 for Cooccurrence. On Spotify MPD, the same comparison is 0.0832 versus 0.0494. Under this metric, larger values mean recommendations are less concentrated in the global popularity head.

Coverage. Coverage also increases modestly. On Last.fm, coverage rises from 0.00166 for Cooccurrence to 0.00175–0.00198 for the LLM conditions. On Spotify MPD, coverage rises from

0.00988 for COOCCURRENCE to 0.01053–0.01194 for the LLM conditions. These gains are small in absolute value, but they indicate that accuracy improvements are not driven only by repeating the most common items.

5 Discussion

What do the results mean? Across both tasks, prompt-only frontier LLMs outperform the implemented classical baselines by large margins. The result is strongest in the MINIMAL condition, which suggests that the models can infer short-horizon continuation cues from recent listening history or playlist context without additional user-summary text. In practical terms, a user who exports listening data and asks a frontier model to rank a fixed candidate set can obtain recommendations that are much stronger than lightweight popularity or cooccurrence heuristics.

Why does the simple prompt work better? The most plausible explanation is that once the candidate set is fixed, the task is mainly about local coherence. Recent tracks provide strong evidence about artist clusters, genre, mood, and sequence flow. Compact profile summaries may dilute that signal by introducing broader but less relevant preferences. The SPOTIFY MPD ablation is especially consistent with this interpretation because the profile prompt is reliably worse for both models.

Why are the gains plausible? The candidate sets are small, and the task rewards semantic matching between the recent context and one correct continuation. Frontier LLMs are likely strong at recognizing style, artist, and playlist-theme compatibility in text form. That capability can dominate lightweight history-based heuristics, especially when the context is coherent. The results therefore support a narrow but credible claim: semantic reasoning over recent history is enough to beat simple public baselines in a reranking setting.

Popularity and diversity. The LLM conditions are not merely replaying the popularity head. They consistently achieve higher mean top-10 popularity percentiles and slightly higher coverage than the classical baselines. This does not prove better novelty in a user-centric sense, but it does show that the gains are compatible with less head-heavy recommendation lists.

Limitations. The paper does not compare against Spotify's production recommender. We evaluate public datasets, a closed 20-item candidate pool, and two lightweight baselines. The LAST.FM experiment required switching from a sparse local subset to a larger remote dataset slice, and the SPOTIFY MPD evaluation uses only one downloaded slice rather than the full benchmark. Harder negative sampling could reduce all scores. Finally, we perform no human evaluation, so the study measures offline ranking accuracy rather than subjective playlist quality or satisfaction.

Broader implications. The main broader implication is about portability rather than raw leaderboard performance. Prompt-only recommendation offers a user-controlled interface to personal history that does not require access to a platform's internal feature store or retraining pipeline. At the same time, our findings should not be overstated. A production recommender typically uses richer feedback signals, larger candidate generation systems, and long-term optimization targets that are absent here. The most realistic path forward may therefore be hybrid: use classical retrieval or sequential models for candidate generation, then use an LLM as a semantic reranker or explanation layer.

6 Conclusion

We studied whether prompt-only frontier LLMs can recommend music from listening history or playlist context without any task-specific training. Using a shared closed-set ranking protocol on LAST.FM and SPOTIFY MPD, we found that both GPT-4.1 and CLAUDE SONNET 4.5 outperform lightweight classical baselines by large margins. The best system, GPT-4.1 with a minimal context prompt, reaches 0.589 MRR on LAST.FM and 0.329 MRR on SPOTIFY MPD, well above the strongest implemented baseline.

The key takeaway is simple: prompt-only LLM recommendation can work surprisingly well against lightweight public baselines, especially when the task is framed as reranking a fixed candidate pool from recent context. The richer profile prompt does not help, and the two frontier models are statistically tied in the strongest setting.

The main next steps are clear. Future work should compare against stronger public baselines such as SASRec, PureSVD, or RecBole models; separate candidate generation from LLM reranking; evaluate real personal Spotify exports; and add human judgments of novelty, usefulness, and playlist fit. Those steps are needed before making any stronger claim about performance relative to production music recommenders.

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